Network Change Report

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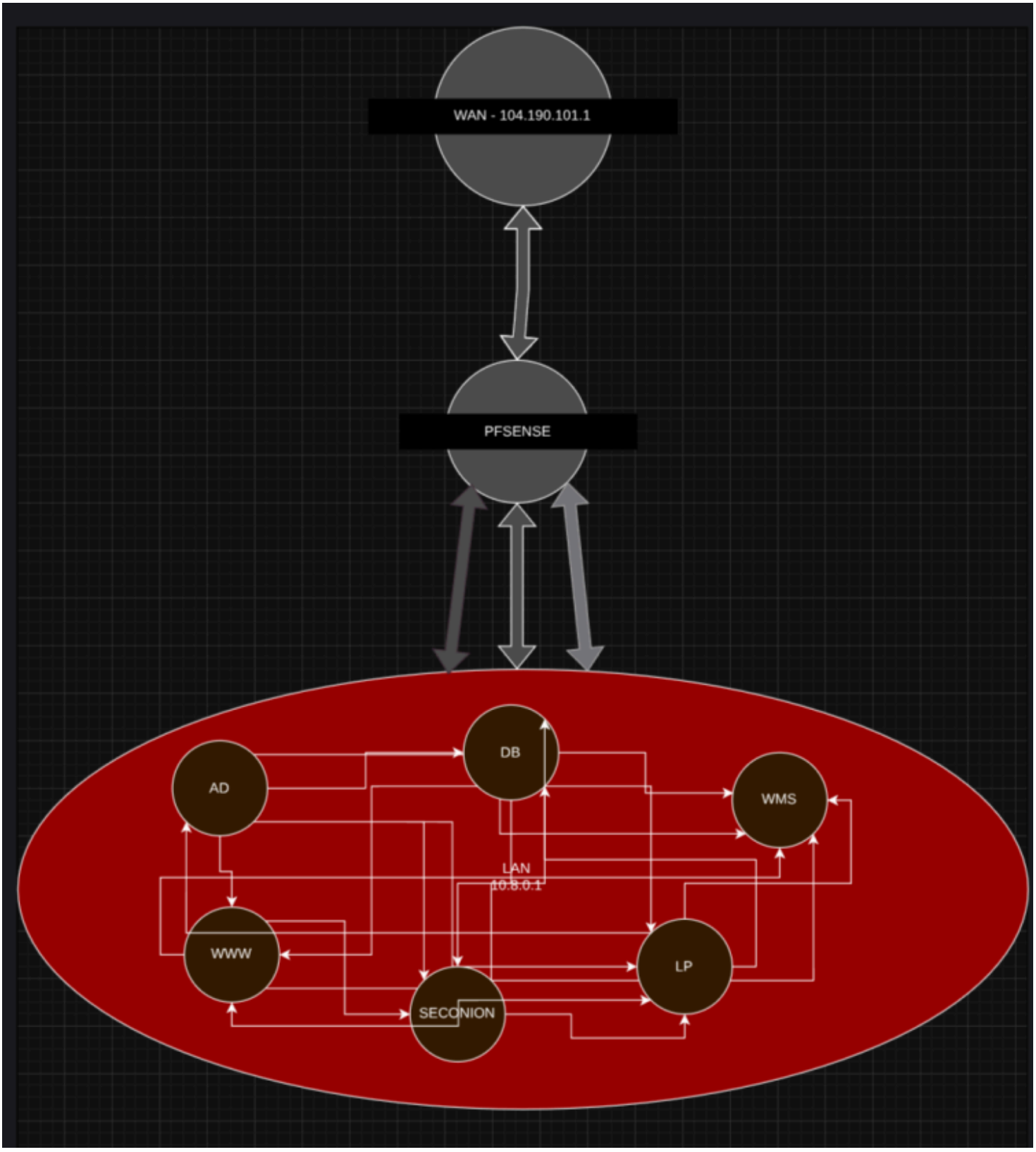
DNS Configuration

Network Layout

Rationale: pfSense acts as the perimeter firewall between the ISEAGE competition network and all internal servers. All scenario servers reside on the LAN segment behind pfSense (`10.8.0.0/24`). Security Onion provides IDS/IPS monitoring on the LAN segment to detect and alert on malicious activity while isolating internal traffic from direct external exposure.


WAN - 104.190.101.1
PFSENSE
DB
AD
WMS
LAN
10.8.0.1
WWW
LP
SECONION

Existing Servers

Host-name Prefix	IP (Local)	IP (Public/WAN)	OS
`AD`	`10.8.0.10`	`104.190.101.10`	Windows Server
`WWW`	`10.8.0.20`	`104.190.101.20`	Alma Linux 8
`DB`	`10.8.0.40`	`104.190.101.30`	Ubuntu 16.04
`WMS`	`10.8.0.30`	`104.190.101.40`	Ubuntu 24.04
`LP`	`10.8.0.50`	`104.190.101.50`	Windows Server 2019

Shared Hardening (Linux Servers)

Hardening and securing steps shared across all Linux systems

Systems covered under this section:

- WWW (World Wide Web)
- DB (Database)
- WMS (Warehouse Management Server)

User Management

Target	Action	Rationale
root	Locked & Removed password	Secured a standard system account to prevent accidental or malicious use.
nobody	Locked	Secured a standard system account to prevent accidental or malicious use.
scrat	Deleted	Removed an unauthorized or unnecessary user account.
sid.john.leguizamo	Deleted	Removed an unauthorized or unnecessary user account. A PAM back-door granted this user perfect `root` permissions even if the shadow file had shown a non-zero UID.

- ...and across all machines, *including Windows Servers,* we have of course, also changed the default credentials for the `cdc` user (Linux) and `Administrator` users (Windows) to complex strings that also remain type-able with moderate effort.

Software Net Positives

- Only this specific section applies to ALL machines, meaning including the Windows machines.

Target	Action	Rationale
All software	Updates	Up-to-date software ensures the latest patches whether that be quality-of-life features, security features, or security patches to counter CVEs.
Bloat-ware (CUPS, Cortana, etc.)	Removed	While not necessarily harmful, poses a small amount of risk and as they're not needed, have been removed to decrease attack surface.

Software Configuration

Logging (AuditD)

Installed the `AuditD` package on every Linux machine and inserted the AuditD rules below @ File `/etc/audit/auditd.rules`:

```
## Monitor all executions
-a always,exit -F arch=b64 -S execve -k execution_monitor
-a always,exit -F arch=b32 -S execve -k execution_monitor

## Monitor file modifications
-a always,exit -F arch=b64 -S write,rename,chmod,chown -k file_mods
-a always,exit -F arch=b32 -S write,rename,chmod,chown -k file_mods

## Lock the configuration
-e 2
```

Logs are to be forwarded remotely to the SecOnion server.

PAM (Pluggable Authentication Module)

Configuration	Action	Rationale
`nullok/nullok_secure`	Removed	The recovery of a locked account is a much better issue than risking a compromise because of a configuration value that says "Empty password? C'mon in!".
`faillock`	Enabled	Standard procedure to lock out accounts after some amount of failures.
`pwhistory`	Enabled	Standard procedure to ensure passwords are not recycled.

SSH (Secure Shell)

Configuration	Old Value	New Value	Rationale
PermitRootLogin	yes	no	Accountability should be clear and actions should be able to be traced back to the responsible person, and therefore direct root login has been disabled.
PubkeyAuthentication	yes	no	Remains un-utilized by the CDC corporation (even though it would be better for security) and so to avoid possible planted hidden keys, has been disabled.
GSSAPIAuthentication	yes	no	Disables unused SSO technology to minimize potential vulnerabilities.
IgnoreRhosts	no	yes	Prevents potential backdoors via user-controlled rHosts files.
IgnoreUserKnownHosts	no	yes	Limits the lateral movement risk if a user account is compromised.
X11Forwarding	yes	no	Disables unused graphical forwarding to reduce the attack surface.
Match User cdc	N/A	`AllowUsers cdc@199.100.16.101`	The `cdc` user is to neve be accessed by anyone other than Blue Team, us. As a result, SSH access to it has been restricted to the Blue VPN.
Subsystem	`SFTP`	None	The `SFTP` subsystem of SSH is not to be utilized as it is not required and poses risk that would otherwise be less likely in the given restricted user-land curated by blue team.

Fail2Ban

Installed and configured for an 8 hour ban on 5 failed authentication attempts in 10 minutes.

- Excluded localhost @ `127.0.0.1/8`
- Excluded the LAN at @ `10.8.0.0/24

Services being monitored:

- SSH/SSHD
- MySQL

Firewall

Firewall Software: `FirewallD`

- Uninstalled any occurrences of the `UFW` in favor of `FirewallD` for its granularity.

Direction	Port	Type	Allowed	Rationale
In	ALL	ALL	No	Default Deny: Blocks all traffic not explicitly permitted.
In	80	TCP	Yes	Required for public access to the storefront and the backend.

Direction	Port	Type	Allowed	Rationale
In	22	TCP	Yes	Required for administrative SSH access.
In	ALL	ICMP Timestamp	No	Prevents reconnaissance attempts that use clock data to identify OS/uptime.
In	ALL	ICMP Echo	No	Disables ping responses to obscure the server's presence from scanners.

System-specific services

Services covered under this section:

- WWW/Front-end
- DB/MySQL Database
- WMS/Back-end

Installed custom SystemD hardening override files to the service's respective overrides folder - The referenced directory can be found at `/etc/systemd/system/(SERVICE NAME).service.d/`, and in that referenced directory, the referenced custom override file is named `override.conf` - The referenced `override.conf` file contains restrictions to reduce the blast radius of compromised services

- Examples of this are found below:
  - `ProtectSystem` (ensures core parts of the system are mounted as read-only for the service)
  - `ProtectHome` (ensures user home directories are not to be read or tampered with) - `NoNewPrivieges` (ensures the service may not re-escalate to previously blocked permissions)
  - Too many to list here. Reference the example override for a generic front-end service below.

```
[Service]
# De-escalate service.
User=www-data
Group=www-data

# Resource limits | Adjust as needed for your scale.
RestrictRealtime=true
TasksMax=64
MemoryMax=2G
MemoryHigh=1800M

# Filesystem protections | Adjust as needed for your setup.
UMask=077
RuntimeDirectoryMode=755
WorkingDirectory=
ProtectSystem=strict
ProtectHome=tmpfs
ReadWritePaths=
BindReadWritePaths=
ReadOnlyPaths=
BindReadOnlyPaths=
NoExecPaths=/tmp /var/tmp /dev/shm

# Capability limitations | Adjust as needed for your setup.
SystemCallArchitectures=native
SystemCallFilter=@basic-io @file-system @system-service @network-io @process @signal @memlock
```

```
RestrictAddressFamilies=AF_INET AF_INET6 AF_UNIX
CapabilityBoundingSet=CAP_NET_BIND_SERVICE
AmbientCapabilities=CAP_NET_BIND_SERVICE

# System & recon protection | Likely won't need to change
ProtectProc=invisible
NoNewPrivileges=true
LockPersonality=true
RestrictNamespaces=true
RestrictSUIDSGID=true
ProtectControlGroups=true
ProtectKernelTunables=true
ProtectKernelModules=true
ProtectKernelLogs=true
ProtectClock=true
ProtectHostname=true
PrivateTmp=true
PrivateDevices=true
DevicePolicy=closed

# May break some applications
PrivateUsers=true

# If using a local database via IPC, change to false.
RemoveIPC=true
```

Aside from applying strict kernel-level restrictions, we also de-escalated all primary services from the `root` user down to a more reasonably permission-ed user such as the commonly used `www-data` user or the `MySQL` user.

Sudoers Configuration

Previous effect	Current effect	Rationale
Allowed all `domain users` to use sudo on all commands	Strictly only allows `it administrators` and the `wheel` group (near-universal agreed upon local administrative group) to use sudo on all commands.	Non-administrators should not be able to conduct changes on a system befitting of administrators.

Software/File removals

File/program	Extent	Rationale
`/root/.john-leguizamo`	Complete purge	Contains unsafe red team source code for backdoor planting. Was thoroughly audited and reverse-engineered to find lingering back-doors
`/lib/x86_64-linux-gnu/libnss_backdoor.so.2`	Complete purge	A PAM standalone C binary meant for use in PAM to intercept credentials. This backdoor likely is a result of the source code found above.
`/usr/bin/john`	Complete purge	The file was a program that had the `SUID` bit, meaning the owner of the file (`root`) was the user that the file would run as. The file ran arbitrary scripts that would compromise the security of the system.

File/program	Extent	Rationale
`C/C++` Compilers	Complete purge	`C/C++` binaries do not need to be compiled aboard the servers. Compilers allow for foreign binaries to be made on-system.
`/etc/crontab`	Singular line removal	The line removed was a `crontab` job that ran as root and occasionally but frequently reset the password of the root user to `cdc`.
GUI/Desktop Environment Packages	Complete purge	As convenient as it is for sysadmins unfamiliar with the terminal, all users managing the Linux systems on our team are familiar somewhat with the terminal and therefore the GUI and any dependencies pose necessary attack surface, such as `polkit`.
`Vi/Vim/Ed`	Complete purge	While the default text editor for many systems, is rather unbounded in its capabilities such as launching new shells and is therefore unsafe for our purposes.
Git-related binaries	Complete purge	Remains unutilized and is removed to reduce attack surface.
`UFW`	Complete purge	`FirewallD` remains the favored firewall for this blue team.
`PAM`	Reinstall	Parts of `PAM` failed system package manager integrity checks and so it was re-installed to remove possible back-doors.

File-system Ownership & Permission Repairs

Applied fixes to the systems' file/directory permission and ownership as per the script below.

```
# Add Sticky Bit to world-writable directories
find / -xdev -type d -perm -0002 ! -perm -1000 -exec chmod +t {} +
#
# For files with an invalid owning user or group, change the owning user & group to root
find / -xdev \( -nouser -o -nogroup \) -exec chmod 640 {} + -exec chown root:root {} +
#
# Remove broken symlinks
find / -xdev -xtype l -exec rm {} +
#
# Ensure sticky-bit on world-writable dirs
chmod +t /tmp /var/tmp /dev/shm
#
# Fix permissions for files regarding identity management
chown root:root /etc/passwd /etc/group /etc/sudoers
if
    grep -q '^shadow:' /etc/group
then
    chown root:shadow /etc/shadow /etc/gshadow
    chmod 640 /etc/shadow /etc/gshadow
else
    chown root:root /etc/shadow /etc/gshadow
    chmod 600 /etc/shadow /etc/gshadow
```

```
fi
chmod 644 /etc/passwd /etc/group
#
# Ensure only root can read the bootloader config
find /boot -type f -exec chown root:root {} + -exec chmod 640 {} +
find /boot -type d -exec chown root:root {} + -exec chmod 750 {} +
#
# Ensure SystemD unit files are secure
find /etc/systemd/system -type f -exec chown root:root {} + -exec chmod 640 {} +
find /etc/systemd/system -type d -exec chown root:root {} + -exec chmod 750 {} +
#
# Secure cron tabs & directories
chown root:root /etc/cron* /etc/at.allow
chmod -R 750 /etc/cron.* /etc/at.allow
chmod -R 640 /etc/crontab
#
# Secure sudoers configuration
chown -R root:root /etc/sudoers /etc/sudoers.d
chmod -R 640 /etc/sudoers
chmod -R 750 /etc/sudoers.d
chmod -R 640 /etc/sudoers.d
#
# Restrict dmesg access
chown -R root:root /bin/dmesg /usr/bin/dmesg
chmod -R 700 /bin/dmesg /usr/bin/dmesg
#
# Secure SSH configurations
find /etc/ssh -type f -exec chown root:root {} + -exec chmod 600 {} +
find /etc/ssh -type d -exec chown root:root {} + -exec chmod 700 {} +
chmod -R 644 /etc/ssh/*.pub
#
# Secure MOTD/banners are secured
chown -R root:root /etc/issue /etc/issue.net /etc/motd
chmod -R 644 /etc/issue /etc/issue.net /etc/motd
#
# Ensure log files are secured
chown -R root:root /var/log
chmod -R 750 /var/log
#
# Secure rsyslog or syslog-ng configs
chown -R root:root /etc/rsyslog.conf /etc/rsyslog.d/*
chmod -R 640 /etc/rsyslog.conf /etc/rsyslog.d/*
#
# Secure Auditd logs & configs
find /etc/audit -type f -exec chown root:root {} + -exec chmod 640 {} +
find /etc/audit -type d -exec chown root:root {} + -exec chmod 750 {} +
#
# Secure global shell profiles
chown -R root:root /etc/profile /etc/bashrc /etc/bash.bashrc /etc/profile.d/*
chmod -R 644 /etc/profile /etc/bashrc /etc/bash.bashrc /etc/profile.d/*
#
# Secure existing home directories
chown -R root:root /root
chmod -R 700 /home/* /root
```

The permissions on the home directories were especially the most offensive, being `755` instead of `700`, meaning anyone could read and execute any files inside another person's home directory.

We also don't get why the default permissions for the `/etc/systemd/system` directory is 755 even though no unprivileged user needs to read from the services nor their override files. We have rectified that to `600` for files in the directory and `700` for directories in the directory.

Extra preventative and noise-making measures

We installed 4 primary custom scripts:

- A script to log all user terminals with the exception of the `cdc` user.
- A script to make the logs from the above immutable once the log is closed and finished.
- A script to lock files in a specific state of permissions and ownership.
- A custom file @ `/etc/profile.d/secure-env.sh` that severely restricts the user-land area/the shell
    - Examples consist but are not limited to the below:
    - Users may not change variables starting with `SSH_`*
    - Users may not change the `USER` or `LOGNAME` variable
    - Users may not reconfigure their shell options (`shellopts`) using `shopt` or `enable`.
    - Users may not use the built-in echo command and must call the external binary
- A custom file @ /opt/mini-bull.sh that

A side-effect of the logging is that non-interactive SSH sessions will not work.

WWW (Public Website)

Software Configuration

Front-end Service

The front-end service formerly granted administrative permissions to all users with a sufficient amount of loyalty points; which has been deemed too much of a security risk and has been removed from the codebase.

There was a function default that defaulted to elevating a user to `Admin` on the front-end/back-end if the function was called without a specified elevation type

We migrated the service's codebase location from the `cdc` user's home directory to the `www-data` user's home directory @ `/home/www-data/`.

We migrated all secrets to the service's SystemD override file.

We migrated the service from plain `Flask` to `Gunicorn` to allow for better resource efficiency and more detailed logs.

Software/File Removals

File/program	Extent	Rationale
`/home/cdc/.bashrc`	Removed the below lines	Disables SELinux and made the `/etc/shadow` file world-writable & world-readable. Clearly undesirable behavior, and may trick users who log on for the first time. (Red team's playing dirty, eh?)

```
sudo chmod -R 777 /etc/shadow
sudo setenforce 0
```

WMS (Warehouse Management)

To comply with the requirement that "...Warehouse Workers must have access to update stock information" swiftly (as the scenario document was most unhelpful in determining in what method to do so), we have decided to allow Warehouse Workers into SSH for WMS with the restriction that they are immediately dropped unto SQL with only the permissions inferred to meet the requirement.

DB (Database)

Software Configuration

MySQL

Action	Result	Rationale
Run `mysql_secure_installation` script	Completed	Applies baseline security hardening to the database engine.
Restrict `richard` SQL user to LAN `.30` & `.20`	Done	Reduces possible openings.
Anonymous Users	Removed	Ensures only authenticated users can access database resources.

AD (Active Directory)

User Management

Action	Result	Rationale
Disabled user accounts troy tomson, guest, scrat	Disabled	Accounts that are unknown or insecure are to be deleted or disabled.
Protected all scenario groups from accidental deletion	Enabled	Prevents red team from potentially reducing availability, is also good practice to prevent accidental deletion

Action	Result	Rationale
Removed user John Leguizamo from IT Administrators	Removed	John Leguizamo is not a documented user nor administrator.

Software Configuration

Security Settings

Target	Action	Rationale
Passwords stored with reversible encryption	Disabled	No reason to have this enabled, only makes it easier for red team to get any potential user passwords.

User Permissions

Action	Result	Rationale
“Everyone” permissions to anonymous users	Disabled	Reduced attack surface for anonymous users.
Disallowed group Domain Users from RDP access	Permissions Removed	Standard domain users do not need access to the RDP service, this reduces manuverability for a potential red team breach.
Removed RDP access from users Guest, scrat, and troy tomsonson	Removed	Removing unknown/insecure users reduces potential nonpermitted logon oppurtunity.
Added group IT Administrators to Domain Administrators	Added	Allows for the IT Administrators group to perform administrative tasks.
Removed group Domain Users from group Domain Administrators	Removed	This rule would allow for all users on the domain to have administrative permissions, allowing for red team to perform any action with the breach of one account.
Gave RDP access to IT Administrator group	Added	Required as per the scenario.

Firewall

Action	Result	Rationale
Disabled firewall rule for Cortana Service	Disabled	Reduces attack surface by disabling unneeded rules.
Users cannot add/login with Microsoft accounts	Disabled	Microsoft accounts are not used in this scenario, reduced attack surface
Auto administrator login for recovery console	Disabled	If red team got the machine into recovery mode, they would have access to any administrator command line permissions.

Action	Result	Rationale
Floppy copy access to all files and folders for recovery console	Disabled	similar rationale as to previous, but with a smaller attack surface
Windows remote management firewall rules	Disabled	Remote management is not the same as Remote Desktop, and is not required for the scenario.
Cast to devices firewall rules	Disabled	This machine does not need to cast to other devices, the rules being on only creates security risk.

Software/File removals

Target	Action	Rationale
Plug 'n Play, print spooler, themes, CPDUserSvc_2d96e	Removed	Reduces attack surface by disabling unused services.
Downloading manufacturers custom icons	Disabled	Non-essential service, disabled for a reduced attack surface.
Deleted malicious script found in Downloads	Removed	Script contained user credentials for the Administrator user, allowing automatic logon.

User & Group Security

Action	Target(s)	Rationale
Disabled Account	guest, scrat, troy tomson	Disables accounts that are unused or suspicious.
Removed from Domain Administrators	Domain Users	Enforces the principle of least privilege for standard users.
Access Restricted	RDP	Restricted RDP access exclusively to the IT Administrators group.

System & Service Configuration

Security Policies

Policy	Setting	Rationale
Network Level Authentication	Enabled	Requires users to authenticate before an RDP session is established.
Reversible Encryption	Disabled	Prevents the system from storing passwords in a decryptable format.

Label Printer

Software Configurations

Windows Defender

Target	Action	Rationale
Windows Defender Advanced Threat Protection Service	Enabled	This service is normally set to enabled by default, and only helps with security. Likely disabled by red team.

Firewall

Target	Action	Rationale
Windows Defender Firewall	Enabled	Firewall rules being turned off allows for red team to have a greatly increased attack surface due to open ports.
Firewall rule named ‘`hackr`’	Disabled	This rule was an `Any`/`Any` `Allow` rule, opening all access to the computer without restriction; is now disabled to prevent red team from easily accessing the system.
Allow Narrator, Cortana Cast to device, and Bluetooth	Disabled	These rules do not serve the scenario or the machine, and thus are unnecessary attack surface.
App and Browser control check apps and files	Set to block	During normal scenario activity, the machine should have no reason to be running a web browser or downloading apps, so this should hinder red team from adding known malicious apps to the system.
Users’ ability to increase a process working set	Disabled	Disabled to block manual buffer overflow attacks
Recycle bin full of unknown files	Emptied	Red team could have placed these files in order to use them in an attack later

User Management

Target	Result	Rationale
cdc	Deleted	CDC is a valid user, but only on the Linux machines, its presence here indicated a potential back-door.

Software/File Removals

Target	Result	Rationale
Blue-tooth, AVCTP, and downloaded maps service	Disabled	Same rationale as above, they do not serve any purpose.

System & Service Configuration

Windows Services

Action	Service	Rationale
Disabled	Blue-tooth	Reduces the physical attack surface by disabling unused protocols.
Disabled	Cortana	Removes unneeded assistant features to improve security.
Disabled	Maps/Narrator	Disables background services not required for a print server.

Network Firewall (pfSense)

Firewall Rules

Direction	Source	Destination	Port	Action	Rationale
ALL	Any	Any	Any	Block	**Default Deny**: All traffic is blocked unless matching a rule below.
In (WAN)	Any	Any	22	Allow	Allow SSH access as needed.
In (WAN)	Any	Any	80	Allow	Web traffic for the storefront.
In (WAN)	Any	`10.8.0.10`	3389	Allow	RDP access for AD management.
In (WAN)	Any	`10.8.0.10`	389	Allow	LDAP access for domain services.
Out (LAN)	`10.8.0.0/24`	Any	Any	Allow	Allows internal servers to initiate outbound connections.
Out (LAN)	`10.8.0.0/24`	`199.100.16.100`	3128	Allow	Specific rule for ISEAGE proxy requirements.

DNS Configuration

Host-name	FQDN	Internal IP
AD	ad.team8.isucdc.com	10.8.0.10
WWW	www.team8.isucdc.com	10.8.0.20
WMS	wms.team8.isucdc.com	10.8.0.30
DB	db.team8.isucdc.com	10.8.0.40
LP	lp.team8.isucdc.com	10.8.0.50